



Paul Leger, Ph. D. in Computer Science & Software Engineer

Software Engineer & Associate Professor
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Personal

Name: Paul Leger
Hometown: Coquimbo, Chile
Nationality: Chilean

Summary:

Google Scholar: CrqCdsgAAAAJ
ORCID: 0000-0003-0969-5139
SCOPUS: 36087550600
DBLP: <https://dblp.org/pid/78/7936.html>
Directors/Headships: 6 Positions
Indexed Journals: 37 Publications (32 WoS-JCR, 2 SCOPUS, and 3 Latindex)
Indexed Conferences: 19 Publications (SCOPUS)
Indexed Workshops: 7 Publications (SCOPUS)
Journal Editions: 6 Journals
First/Corresp Author: 24 (represents 38%)
Courses: 56 Courses (49 undergrad and 7 graduate)
Thesis Supervisions: 27 Theses (24 undergraduate and 3 graduate)
Relevant Software: 9 Developments (2 applications, 3 libraries, and 4 languages)
External Projects: 3 with funding



Education

Visiting Researcher, universities across the world:

- Centre for Development of Advanced Computing (C-DAC), India, 2024 (3 weeks).
- University of The Andes, Colombia, 2021 (3 weeks).
- Tokyo Institute of Technology, Japan, 2019 (6 months).
- University of Cauca, Colombia, 2018 (2 months).
- Shibaura Institute of Technology (x 2), Japan, 2013-2014 (4-2 months).
- École des mines de Nantes, France, 2010 (2 months).

Certifications: *a)* "Fundamentals of English Medium Instruction", University of Pennsylvania, 2023, *b)* "Teaching University", Highest level, Catholic University of the North, 2024.

Ph. D. in Computer Science, University of Chile, 2006 - 2012.

- Subject: Opening Up Trace-Based Mechanisms.
- Advisor: Éric Tanter (<https://pleiad.cl/etanter>).

Engineering degree in Computer Science, University of La Serena, 1999 - 2005 (July).

B.Sc. in Computing Science, University of La Serena, 1999 - 2005 (April).

Management Services

Director. Engineering School (<https://eic.ucn.cl>) at Universidad Católica del Norte (Chile). Management of a budget for a four-floor engineering school with 1,000+ students, guiding professors, and a teaching support unit, 2024 - 2026.

Director. Chilean Computer Science Society (SCCC), 2020 - 2022.

Deputy Director. Engineering Systems Doctorate (<https://dsi-en.ucn.cl>) at Universidad Católica del Norte (Chile), 2023 - 2024.

Head of Program Studies. 1) Civil Engineering in Computer Science and 2) Information Technology Engineering at the Universidad Católica del Norte (Chile), over 400 students, 2021 - 2022.

Pragmatics Lab. Director of an international research laboratory (<http://pragmatics-lab.com>), where professors and students from Japan, Colombia, and Chile collaborate, 2019 - present.

Software Development Engineer

As I enjoy working on software engineering, I occasionally work as a freelancer doing software development and as a team manager in software projects:

Macrosoftware. Consulting software engineer, 2020 - present.

G-Tech. Project Manager, a startup mobile application, 2022.

Autopista Central. Consulting engineer, 2017.

Regional Observatory (working market). Software project manager, implementation of an information system that allows users to analyze working market in a region of Chile, 2016 - 2017.

Autopista Central. Consulting engineer, 2011.

Software Development for Research

Research-driven software systems and developer tools, built with production-grade engineering practices where applicable (open-source repos, demos, and reusable libraries). Source: <https://pleger.cl/software>.

PL/SE research with AI: Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). <https://pleger.github.io/PL-SE-research-with-AI>, 2026. [ARTIFICIAL INTELLIGENCE].

SBABM: Java software to simulate buyers in e-commerce like Amazon, AliExpress, and Wish. An agent-based model to simulate the e-commerce behavior between agents and markets. <https://github.com/pragmatics-laboratory/SBABM>, 2023. [APPLICATION].

EMAJs: JavaScript library that implements an Expressive and Modular Activation mechanism for context-oriented programming. <https://github.com/pragmatics-laboratory/EMAJs>, 2022. [LANGUAGE].

DeloreanJS: A back-in-time debugger for JavaScript. This software is strongly developed by Felipe Ruiz (felipe.ruiz@alumnos.ucn.cl) and Guillermo Victorero (guillermo.victorero@alumnos.ucn.cl). You can try it on <https://pleger.cl/sites/deloreanjs>. <https://github.com/pragmatics-laboratory/deloreanjs>, 2019. [APPLICATION].

AspectScript: An aspect-oriented extension of JavaScript for expressive AOP, which integrates several state-of-the-art features related to higher-order programming and expressive scoping of aspects. <https://pleiad.cl/aspectscript>, 2010. [LANGUAGE].

FASOW: FASOW provides an Open Implementation for Agent-Based Models applied to e-WOM Marketing Campaigns. This software is strongly developed by Diego Nuñez (diego.nunez.trabajo@gmail.com). <https://github.com/pragmaticslaboratory/fasow>, 2024. [LIBRARY].

Matcher Cell: A JavaScript library to flexibly match patterns. You can try it on <https://pragmaticslaboratory.github.io/matcher-cells-study-cases>. <https://github.com/pragmaticslaboratory/match-cell-base>, 2021. [LANGUAGE].

Sync/CC (proof of concept): A library to address callback issues on JavaScript. You can try it on <https://pleger.cl/synccc>. <https://github.com/pragmaticslaboratory/synnccc>, 2019. [LIBRARY].

Academic Experience

Scientific Services

Funded Projects (x 4). Principal Researcher (ANID-2023), Principal Investigator (Chile), and Co-Researcher (Japanese government).

Organizer of International Events (x 4). JCC'20 (Chile), COP'20 (Germany), and MPSE'14/15 (USA).

Editor and Reviewer of JCR Journals (x 48). Editor (x 5) and Reviewer (x 43).

Chair of International Events (x 9). ICSME'23 (Colombia), LMN'22 (Chile), JCC'20 (Chile), MPSE'14/15 (USA), SAC-SE'17 (Morocco), and BIONETICS'15 (USA).

Program Committee Member (x 21). VMIL (Singapore), ICSME (various countries), SAC-SE (various countries), SBLP (Brazil), COP (Germany), VISSOFT (Chile), INFONOR (Chile), etc.

Evaluator (x 16). Research Projects and (International) External Master Reviewers.

References

Hidehiko Masuhara. Ph. D. in Computer Science. Full Professor & Dean. Department of Mathematical and Computing Science. Tokyo Institute of Technology, Japan (masuhara@acm.org). Webpage: <https://prg.is.titech.ac.jp/people/masuhara>.

Éric Tanter. Ph. D. in Computer Science. Full Professor. Computer Science Department (DCC), University of Chile, Chile (etanter@dcc.uchile.cl). Webpage: <https://pleiad.cl/people/etanter>.

Alexandre Bergel. Ph. D. in Computer Science. Computer Scientist. Rational AI (<https://relational.ai>), Switzerland (alexandre.bergel@me.com). Webpage: <https://bergel.eu>.